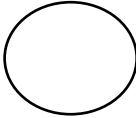
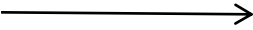
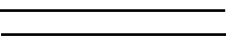


11. Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) below represents the flow, processing, and storage of data within all of the Medical Appointment & Prescription System's processes, providing an in-depth illustration of how the database is organized by the stakeholders.

11.1 DFD Components & Symbols

There are four standard notations used to construct and read a DFD correctly:

Symbol / Element Name	Visual Representation	Description
Process		Used to depict the system or a sub-system that receives input data and converts it into output data.
External Entity (Actor)		Used to depict external entities such as sources or destinations of data, for instance, Admin, Staff, or Doctor.
Data Flow		represents the flow of data between entities, processes, and data stores
Data Store		represents a repository where data is stored for future use (e. g. MySQL tables).

11.2 DFD Level 0 (Context Diagram)

The A0 level Data Flow Diagram known as the Context Diagram is the top-level depiction of the processes within the system. In other words, the whole software application is visualized as one particular process that delivers data to external actors through input and output flows.

The following diagram below represents an elaborated theoretical description of the DFD level 0 for the Medical Appointment & Prescription System.

i Central Core Process (Process 0: Medical Appointment & Prescription System)

- **Gatekeeping & Access Control:** Central platform for processing all interface requests, regulating access to the system by all users through the implementation of access control mechanisms.
- **Scheduling Coordination:** Coordinates and processes all booking requests from users, checking availability against the doctor's schedule while also resolving conflicting requests.
- **Data & Record Management:** Maintains continuous write/read operations to the MySQL database, including the operation of electronic prescriptions, treatment records, and user data.
- **Notification & Output Dispatch:** Manages the execution of all notification tasks and schedules alerts and reminders to system users while also generating PDF summary reports of medical records.

i. External Entities (Actors) & Detailed Description

External Entities (Actors) refer to those sources or destinations of data, which are outside the boundaries of the system under consideration. Following is a 7-point list, describing the key roles played by external entities (actors) in the Medical Appointment & Prescription System:

- **Admin:**

The system administrator who supervises the set up of the whole system including managing and updating all the necessary software in the system. The following are the duties performed by an admin:

- ◆ **Identity & Access Management:** Used to control software account life cycle and clinic staff/doctor directory, while also managing system roles, permissions, and access.
- ◆ **System & Database Administration:** Configuring system's core backend databases and software settings by utilizing administrative master controls to override standard operating procedures.
- ◆ **Security & Compliance Auditing:** Performing log-file level monitoring of the system's activity to detect and alert unauthorized behavior, illegal access, or privilege escalation.

- **Staff:**

Front desk or receptionist who deals directly with all patient information including managing and monitoring all activities relating to patient appointments. The responsibilities entailed when dealing with this function include:

- ◆ **Patient Intake & Information Retrieval:** Entering of patient's contact and demographic information into the system for easy identification and access of their records through the patient's ID or name.
- ◆ **Appointment Scheduling & Conflict Avoidance:** This feature helps avoid scheduling conflicts by ensuring that only 'available' doctors are selected when booking appointments with patients.
- ◆ **Status Monitoring & Point of Contact:** Managing appointment status and point of contact for patients in a day by day basis which includes updating patient records to indicate whether a scheduled appointment was confirmed, cancelled or completed.

- **Doctor:**

This is a medical practitioner who provides medical services to the patient by viewing their medical history or clinical notes, viewing or managing their appointment status and viewing the schedule for a day. The responsibilities performed by this entity include:

- ◆ **Availability & Shift Management:** Sets the core working hours, specific timings, and shifts between available, on break, and unavailable statuses
- ◆ **Queue Monitoring & Patient Consultation:** Views the queue dashboard to respond to medical consultations while accessing the patient's database history.
- ◆ **Clinical Records & Prescription Dispatch:** Scans through clinical histories to authorize electronic prescriptions, clinical consultation records, and the sending or receiving of medical summaries and slips in downloadable PDF format. Now that we

- **Patient:**

This is a person who comes to the clinic to receive services rendered by the doctor. The patients provide information about their personal contacts and demographics during front-desk registration. They also attend to or come to physical consultation at the clinic on appointment days and times. The duties of this entity are:

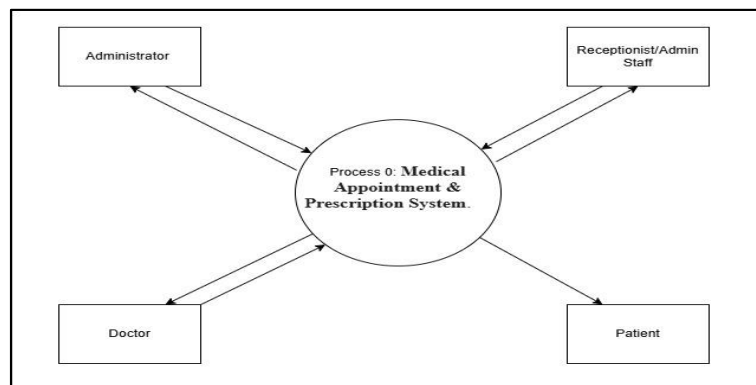
- ◆ **Registration and consultation access:** Provide personal demographics at the front desk and check in at the clinic or through the online portal at the scheduled time for medical consultation
- ◆ **Care communication and digital output:** Consult directly with medical personnel, give feedback about the care received, and get digital medicine slips and treatment reports.
- ◆ **System alerts and notifications:** Receive automated messages about the status of appointments, reminders about visits, and notifications about medicine doses.

ii. Data Flows

DFD level 0 presents the data flow diagram for the Appointment & Prescription System. The notations below explain which of these data flow vectors are inbound (from Actors) and outbound (to Actors) and what data they carry:

- **Inbound :** External entities like the Administrator, Receptionist/Staff, and Doctor provide the user's credentials, patient's information, appointment records, and prescription to the central process.
- **Outbound:** The process provides response to the actors and patients like the appointment schedule confirmation, dashboard access, and medical reports display.

iv Diagram



11.3 DFD Level 1

The Level 1 Data Flow Diagram is an elaborated version of the diagram above it that fully decomposes the central process into a number of lower-level processes and introduces data stores that fully describe the data flows in the information system.

i. Detailed Breakdown of Level 1 Sub-processes

The Level 1 Data Flow Diagram (DFD) below breaks down the core process into six sub-processes that perform distinct operations, involve different actors, and utilize various data storage systems. The description below provides a 3-4 point breakdown of each of the Level 1 sub-processes:

- **1.0 Login (Authentication and Access Control)**
 - ◆ The process serves as the point of entry to the desktop software where all other processes take place.
 - ◆ It reads from and writes into the D1 Users data store to cross-check login credentials and determine the privileges of each user that accesses the dashboard.
 - ◆ The process also regulates the login session and sends error messages to users that provide wrong login details.
- **2.0 Patients (Patient Management)**
 - ◆ The process is used to register and manage all patient information in the system.
 - ◆ It reads patient data and writes into the D2 Patient data store to record new patient information in the system.
 - ◆ It also validates the information provided by patients at registration and issues a unique patient ID for every successfully registered patient.

- **3.0 Doctors (Doctor Directory)**

- ◆ The process allows the administrator to manage the doctor directory, including their departments and fees.
- ◆ It reads doctor information and writes into the D3 Doctors data store to update the doctor's contact information and fees.
- ◆ The process also regulates the doctor's specialization in terms of departments and updates the directory in the D3 Doctors data store.

- **4.0 Appointments (Appointment Management)**

- ◆ The process allows the front desk executive to schedule appointments by selecting preferred doctors and dates while also considering the availability of doctors.
- ◆ It reads appointment information and writes into the D4 Appointment data store to update the appointment status and remind patients of their scheduled visits.
- ◆ The process also tracks the status of appointments, including pending, confirmed, completed, canceled, and other statuses, to manage and update the front desk operations.

- **5.0 Prescriptions**

- ◆ The process allows doctors to manage prescriptions and notes related to medical treatments during patient consultations.
- ◆ It reads prescription information and writes into the D5 Prescription data store to record the medication instructions and treatment notes for patient files.
- ◆ The process tracks medication instructions, including the medicine name, dosage, intake instructions, duration, and other relevant information.

- **6.0 Reports (Reporting, Reminders, and Output)**

- ◆ The process provides reports and reminders to the relevant stakeholders involved in the operation of the medical clinic.
- ◆ It reads information from all data stores, including D2 Patients, D3 Doctors, D4 Appts, and D5 Rx, to generate statistics and reports.
- ◆ It also provides the final medical reports in PDF format to the relevant stakeholders, including the treatment notes and medication slips for patients.

ii. External Entities (Actors) & Detailed Description

In relation to the Level 1 DFD, external entities denote the input and output sources of the information system. Hence, the following is a 3 to 4-point description of each entity:

- **All users (Admin, staff, and doctor)**

- ◆ This entity encompasses every user category and provides the unified entrance to the application. All the users need to authenticate their access to the system using their respective login credentials.
- ◆ This entity offers the input data in terms of private login usernames and strong passwords to the authentication segment.
- ◆ It presents the dashboard interface depending upon the user category and facilitates simultaneous log out options.

- **Staff (receptionist/front-desk)**

- ◆ This entity offers the input data in terms of patient's personal details recorded during front-desk registration.

- ◆ It provides the patient's complete information and available doctor's schedule to the booking segment.
- ◆ The entity also offers the status queue and facilitates fast record lookup in the system
- **Admin (Administrator)**
 - ◆ The entity offers the overall configuration and setup input data to the application.
 - ◆ It provides the user account creation segment with new staff, their credentials, and roles within the system.
 - ◆ The entity also presents the directory management segment with new staff or doctor's professional directory, including the editing and deletion options.
- **Doctor (Medical practitioner)**
 - ◆ This entity offers the doctor's availability segment for schedule setup within the system.
 - ◆ It provides the clinical history view segment and the prescription generation segment with the medication's name, dosage, and additional clinical instructions.
 - ◆ The entity presents the daily status queue view on the doctor's dashboard.
- **Patient (care recipient)**
 - ◆ This entity provides the personal detail input segment at the front-desk or registration segment.
 - ◆ The patient receives the medical slip, printed prescription, and the PDF summary of the treatment.
 - ◆ They also obtain reminder access to the appointment's status and time.

iii. Data Stores & Descriptions

In a Level 1 Data Flow Diagram (DFD), Data Stores are generally represented by places where records are kept, maintained, and stored for later retrieval and use. Below are the 3-to-4-point descriptions of each data store identified in the Medical Appointment & Prescription System Data Flow Diagram

- **D1 Users**
 - ◆ **Description:** This is a data store for user authentication and authorization details such as the credentials and access levels.
 - ◆ **MySQL Table:** Users MySQL table using user_id as they key
 - ◆ **Details:** This data store holds encrypted passwords, usernames, and user roles thus controlling access to the Medical Appointment & Prescription System.
- **D2 Patients**
 - ◆ **Description:** This is a data store for the personal information of all patients registering for medical services.
 - ◆ **MySQL Table:** Patients MySQL table using patient_id as the key
 - ◆ **Details:** This data store holds the personal contact and demographic information of all patients.
- **D3 Doctors**
 - ◆ **Description:** This is a data store for the personal information and professional credentials of all doctors offering medical services at the facility.
 - ◆ **MySQL Table:** Doctors MySQL table using doctor_id as the main key and user_id as a foreign key.
 - ◆ **Details:** This data store holds the professional qualifications, specialization, and fees charged by the doctors as well as their schedules.

- **D4 Appointments**

- ◆ **Description:** This is a data store for holding information about the appointments including the patient, the doctor, the date, and time of the scheduled visits.
- ◆ **MySQL Table:** Appointments MySQL table using appointment_id as the main key with patient_id and doctor_id as foreign keys.
- ◆ **Details:** This data store records the appointment dates and times as well as the status of the appointments.

- **D5 Prescriptions**

- ◆ **Description:** This is a data store for electronically capturing the doctor's prescriptions such as the medication instructions and treatment notes.
- ◆ **MySQL Table:** Prescriptions MySQL table using prescription_id as the main key with foreign keys appointment_id and doctor_id.
- ◆ **Details:** This data store holds the medication instructions including the type of medication, frequency and duration of dosage, and additional treatment notes.

iv. Data Flows

In a Level 1 Data Flow Diagram (DFD), the data flows are categorized according to the process they originate from. The diagram below contains a 3-4 point detailed description of the inbound and outbound data flow vectors of each process in the Medical Appointment & Prescription System:

- **1.0 Login (Authentication Module) – Data Flow Vectors**

- ◆ **Inbound: (Actor → Process):** All actors enter their login credentials and clear text passwords into a given Authentication Module window.
- ◆ **Bidirectional: (Process → Data Store):** Run read/write query commands with the D1 Users data store in order to extract the stored hash(password) and roles.
- ◆ **Outbound: (Process → Actor):** Grant Access and redirect the authenticated actor to the respective role's dashboard; alternatively, output errors if the authentication attempt was unsuccessful.

- **2.0 Patients (Patient Management Module) – Data Flow Vectors**

- ◆ **Inbound: (Actor → Process):** Front-desk Staff members input all the requisite data necessary to onboard a new patient and any search queries.
- ◆ **Bidirectional: (Process → Data Store):** Run read/write commands with the D2 Patients data store to write new data records and extract past patient data.
- ◆ **Outbound: (Process → Actor):** Output success notices, the generated patient_id, or the relevant inline error messages.

- **3.0 Doctors (Doctor Directory & Profile Module) – Data Flow Vectors**

- ◆ **Inbound: (Actor → Process):** The Admin inputs the Doctor's medical practice area, professional fees, and relevant specializations with the Doctor Directory & Profile Module.
- ◆ **Bidirectional: (Process → Data Store):** Run read/write commands with the D3 Doctors data store in order to onboard new doctors with authentication tokens (user_id).
- ◆ **Outbound: (Process → Actor):** Output the administrative success notice, updates made to the doctor directory, and the full list of doctors with relevant specialties, along with contact information.

- **4.0 Appointments (Appointment Booking Module) – Data Flow Vectors**

- ◆ **Inbound: (Actor → Process):** Front-desk Staff and Doctors input appointment slots, availability, and booking requests.
- ◆ **Bidirectional: (Process → Data Store):** Run read/write commands with the D4 Appts data store in order to process, store, and update the appointment queue database.
- ◆ **Outbound: (Process → Actor):** Output confirmation of appointment requests, alert staff/doctors regarding any clashes, and the updated queue status.

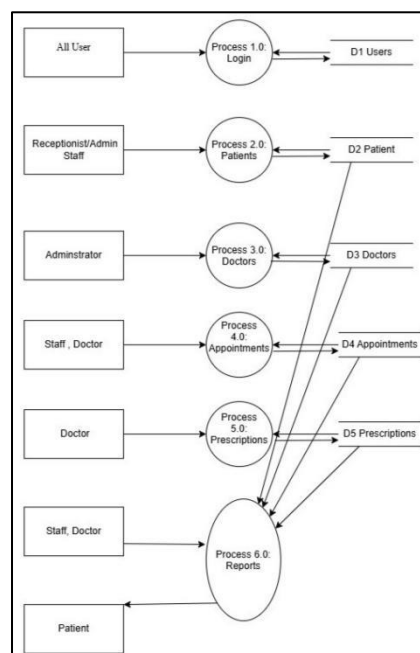
- **5.0 Prescription (Prescription Service Module) – Data Flow Vectors**

- ◆ **Inbound: (Actor → Process):** The Doctor inputs the newly issued prescription, medication instructions, and remarks into the Rx Prescription Service Module.
- ◆ **Bidirectional: (Process → Data Store):** Run read/write commands with the D5 Rx data store to write any new prescription data records as well as extract any relevant medication instructions from the past data.
- ◆ **Outbound: (Process → Actor):** Output verification of submitted prescriptions, preview digital prescriptions, inline error messages, and downstream data feeds.

- **6.0 Reports (Reporting, Reminders, and Output Module) – Data Flow Vectors**

- ◆ **Inbound: (Data Store → Process):** Receive inbound data directly from D2 Patients, D3 Doctors, D4 Appts, D5 Rx containing operational data statistics.
- ◆ **Inbound: (Actor → Process):** Staff and Doctors input their export and reporting requests.
- ◆ **Outbound: (Process → Actor):** Output PDF medical summary reports to Patients, appointment reminders to Staff and Doctors, and finally, end-of-day logs and reminders.

v. Diagram



11.4 DFD Level 2

A Level 2 Data Flow Diagram (DFD) is a context diagram that performs the expansion of a process listed in a Level 1 DFD into sub-processes. A Level 2 DFD is the detailed version of a process and illustrates the internal workings of activities performed at a data flow in a system.

i. Detailed Breakdown of Level 2 Sub-Processes

The Level 2 Data Flow Diagrams have been developed in order to provide further insight into the internal logic and data flows pertaining to certain Level 1 modules. Each Level 2 process contains detailed steps that specify exactly how the identified business functions and processes have to be performed in order to yield the required results. In accordance to the aforementioned objective, the subsequent text provides a comprehensive ‘three to four points’ breakdown of all Level 2 processes within the Medical Appointment & Prescription System Data Flow Diagram below:

- **Step-by-step breakdown of process 4.0**

- ◆ **4.1 Booking & 4.2 Check Avail.:** This process is triggered by either external Staff or Doctor inputs in order to set up a medical appointment. It subsequently accesses data within the D3 Doctors data store in order to ensure that the selected doctor is indeed available throughout the proposed booking period.
- ◆ **4.3 Clash Check & 4.4 Save Booking:** The process then queries the D4 Appointments data store in order to perform an additional booking clash check. After the relevant booking details have been validated against existing records, the validated information is then written to the D4 Appointments data store.
- ◆ **4.5 Set Status:** The process updates the relevant booking status (e.g. pending) details within the D4 Appointments data store and finally provides feedback to both the Staff and Doctor in the form of output.

- **Step-by-step breakdown of process 5.0**

- ◆ **5.1 Enter Rx & 5.2 Validate:** This process is performed by Doctors whenever they wish to electronically log a new prescription against their corresponding patient’s file. It subsequently validates the entered prescription data against the provided clinical notes in order to ensure that complete data is captured and stored within the system.
- ◆ **5.3 Save Prescription:** The processed digital copy of the new prescription is then written to the D5 Prescriptions data store.
- ◆ **5.4 Gen. Reminder & 5.5 Send Reminder:** The process then accesses relevant prescription information from the D5 Prescriptions data store in order to generate and send out electronic patient reminder messages. The finalized electronic reminder is thereafter sent to the corresponding patient.

ii. External Entities (Actors) Description for Level 2 Workflows

When it comes to external entities (actors) related to Level 2 DFD workflows (Appointment Scheduling and Prescription Management), they should be identified as the particular initiators or recipients of data from the process steps specified at the sub-process level. Below are the 3-4 itemized descriptions of the entities related to the outlined below Level 2 processes:

External Entities Description

- **Staff (receptionist/front-desk personnel)**

- ◆ **Appointment initiation process:** The sub-process 4.1 Booking is initiated by staff by entering the required patient and selected doctor’s details.
- ◆ **Schedule coordination:** As the first party involved in consult request processing, staff is accountable for coordinating the consult schedule in accordance with the availability of doctors.
- ◆ **Workflow confirmation notificatory:** Once the booking sub-process is executed and the confirmation is stored in the database, the staff receives the status confirmation notificatory.

- **Doctor (medical practitioner)**

- ◆ **Schedule and appointment input:** This process participant is involved in either appointment scheduling or prescription management process and initiates 5.1 Enter Rx by providing the necessary treatment-related details.
 - ◆ **Treatment parameters provision:** In terms of treatment entry, a doctor is considered the provider of prescription-related data, including drug name, frequency, dose, additional clinical instructions.
 - ◆ **Workflow facilitator:** As a medical practitioner, a doctor facilitates the consult and related prescription management workflow steps and makes sure that consult input is integrated into the system in due course.
- **Patient (care recipient)**
 - ◆ **Reminder target:** In relation to the reminder sub-process (5.5 Send Reminder), the patient is the end recipient of the automated reminder.
 - ◆ **Treatment adherence enabler:** As a care recipient, the patient is the one who benefits from the treatment reminder notifications delivered via the alert channel.
 - ◆ **Outbound notificatory beneficiary:** From the perspective of notifying about an upcoming clinical interaction, the patient is regarded as the end beneficiary of the corresponding outbound notificatory.

iii. Data Stores & Detailed Descriptions (Level 2 Workflows)

In the Level 2 Data Flow Diagrams, the data stores engage in extensive record-keeping of lookup activities, validations, and state transitions. Below are the detailed 3-4 item descriptions per relevant data store used in the process:

- **D3 Doctors (Professional Profiles Data Store)**
 - ◆ **What it does best:** This is the record-keeping database which maintains doctors' schedules for the purpose of availability checking.
 - ◆ **Read access by sub-process 4.2 Check Avail.:** It is accessed for information on working hours and general availability of the professional under consideration.
 - ◆ **Directory capabilities for sub-process 4.1 Make Request:** It provides the necessary information on departments and scheduling availability for consultation purposes.
- **D4 Appointment (Appointment Scheduling Data Store)**
 - ◆ **What it does best:** This data store is used for the actual transaction of booking a medical consultation as well as maintaining its current state.
 - ◆ **Clash checking by 4.3 Clash Check:** It is queried for the purpose of determining whether a new appointment request would involve any time clashes.
 - ◆ **Recording process for 4.4 Save Booking:** It is updated with the relevant information on the consultation saved.
 - ◆ **State transition recording by 4.5 Set Status:** It receives updates on the current state of an appointment, e.g. whether it is pending or confirmed.
- **D5 Prescriptions (Prescription & Treatment Data Store)**
 - ◆ **What it does best:** This database is used to keep electronic records of all prescriptions and treatments provided by the facility.
 - ◆ **Recording treatment by sub-process 5.3 Save Prescription:** It is updated with the relevant information on the new treatment recorded.
 - ◆ **Reminder generation by 5.4 Gen. Reminder:** It is queried for the necessary information on treatment schedules and dosages which would be needed to set up reminders for patients.

iv. Detailed Data Flows (Inbound/Outbound Vectors by Level 2 Process)

In the Level 2 Data Flow Diagram, the process is decomposed into its component sub-processes and data flows that illustrate the exchange of information in more detail. In each Level 2 DFD, the data flows are represented between actors, subsequent sub-processes, and persistent data stores. Below are the 3-4 point descriptions of inbound (in) and outbound (out) information vectors for each of the Level 2 processes:

- **4.0 Appointment Scheduling Vectors**

- ◆ **4.1 Booking & 4.2 Check Avail. (in):** Staff or Doctor initiates the process by entering booking details, while sub-process 4.2 reads the doctor's working hours and status information from the D3 Doctors data store.
- ◆ **4.3 Clash Check & 4.4 Save Booking (in/out):** Sub-process 4.3 checks availability by querying the D4 Appointments data store for clashes, while sub-process 4.4 writes and stores booking details in the D4 Appointments
- ◆ **4.5 Set Status (out/in):** It updates the active status information in D4 Appointments and sends back booking confirmation messages to the Staff and Doctor.

- **5.0 Prescription & Reminder Vectors**

- ◆ **5.1 Enter Prescription & 5.2 Validate (in):** Doctor enters the patient's prescription and relevant clinical information that is then validated.
- ◆ **5.3 Save Prescription (in):** It writes the digital prescription information into the D5 Prescription data store.
- ◆ **5.4 Gen. Reminder & 5.5 Send Reminder (in/out):** Sub-process 5.4 reads the treatment details from D5 Prescription data store and sub-process 5.5 sends out the digital reminder to the Patient.

v. Diagram

